
Phase 8: Project Demonstration

SmartLender AI-Powered Credit Card Approval System



Demo Scenario 1 – Automated Screening

A credit analyst enters an applicant's financial profile into the web application. The application returns an instant approve/reject prediction, letting the analyst prioritize which applications need closer review.

Demo Scenario 2 – High-Risk Identification

A compliance officer screens applicants with past-due loan records. The feature engineering pipeline converts multi-class payment status into a binary risk score. High-risk applicants are clearly flagged.

Demo Scenario 3 – Self-Service Eligibility Check

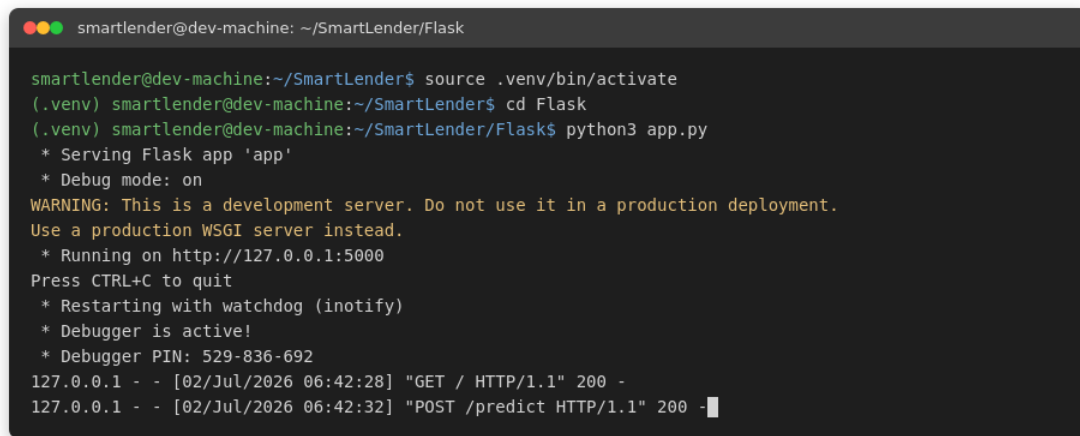
A prospective customer enters their own details before formally applying and receives an instant likelihood of approval, reducing the number of applications that would otherwise be rejected outright.

Conclusion & Future Scope

SmartLender shows that a tuned XGBoost model can automate a meaningful share of the credit approval workload. Future work includes adding explainability for each decision and monitoring the model for drift as new applicant data comes in.

Terminal Execution

The screenshot below shows the Flask development server starting successfully and serving requests on local port 5000.

A terminal window with a dark background and light text. The prompt is 'smartlender@dev-machine: ~/SmartLender/Flask'. The user enters 'source .venv/bin/activate', then '(.venv) smartlender@dev-machine:~/SmartLender\$ cd Flask', and finally '(.venv) smartlender@dev-machine:~/SmartLender/Flask\$ python3 app.py'. The output shows the server starting, including a warning about production use, the port (5000), and a debugger PIN. Two HTTP requests are shown: a GET request to '/' and a POST request to '/predict', both returning 200.

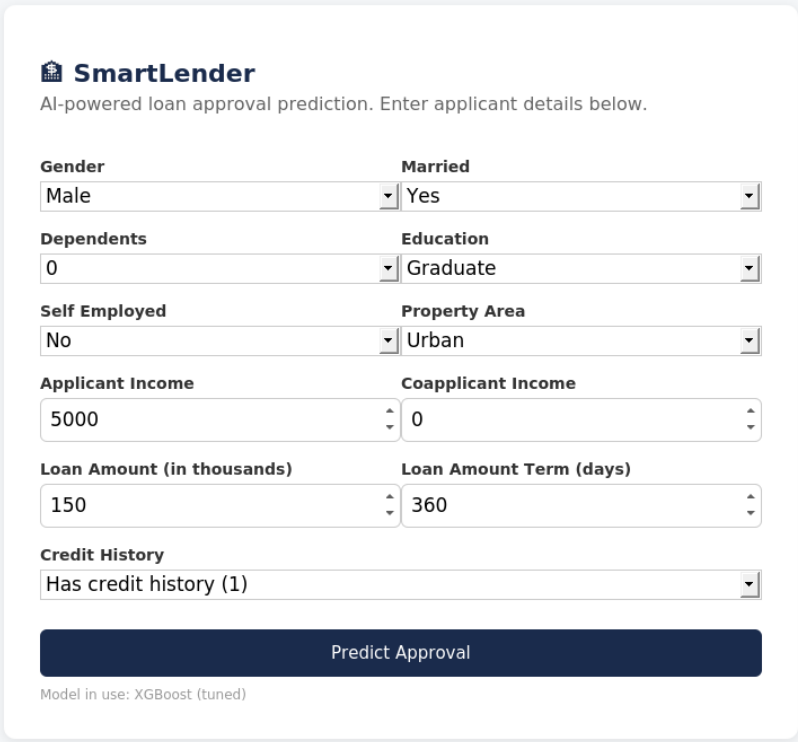
```
smartlender@dev-machine: ~/SmartLender/Flask

smartlender@dev-machine:~/SmartLender$ source .venv/bin/activate
(.venv) smartlender@dev-machine:~/SmartLender$ cd Flask
(.venv) smartlender@dev-machine:~/SmartLender/Flask$ python3 app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (inotify)
* Debugger is active!
* Debugger PIN: 529-836-692
127.0.0.1 - - [02/Jul/2026 06:42:28] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Jul/2026 06:42:32] "POST /predict HTTP/1.1" 200 -
```

Figure 5: Flask server running in terminal

Web Interface – Input Form

The applicant enters demographic and financial details through the web form shown below.



The image shows a web interface for 'SmartLender', an AI-powered loan approval prediction tool. The interface is a light gray card with a white background, set against a light blue background. At the top left is the 'SmartLender' logo, which consists of a small house icon with a dollar sign inside, followed by the text 'SmartLender'. Below the logo is the text 'AI-powered loan approval prediction. Enter applicant details below.' The form contains several input fields arranged in two columns. The first column includes 'Gender' (a dropdown menu with 'Male' selected), 'Dependents' (a dropdown menu with '0' selected), 'Self Employed' (a dropdown menu with 'No' selected), 'Applicant Income' (a text input field with '5000' entered), 'Loan Amount (in thousands)' (a text input field with '150' entered), and 'Credit History' (a dropdown menu with 'Has credit history (1)' selected). The second column includes 'Married' (a dropdown menu with 'Yes' selected), 'Education' (a dropdown menu with 'Graduate' selected), 'Property Area' (a dropdown menu with 'Urban' selected), 'Coapplicant Income' (a text input field with '0' entered), and 'Loan Amount Term (days)' (a text input field with '360' entered). At the bottom of the form is a large, dark blue button with the text 'Predict Approval' in white. Below the button, in small gray text, is the note 'Model in use: XGBoost (tuned)'.

SmartLender
AI-powered loan approval prediction. Enter applicant details below.

Gender Male	Married Yes
Dependents 0	Education Graduate
Self Employed No	Property Area Urban
Applicant Income 5000	Coapplicant Income 0
Loan Amount (in thousands) 150	Loan Amount Term (days) 360
Credit History Has credit history (1)	

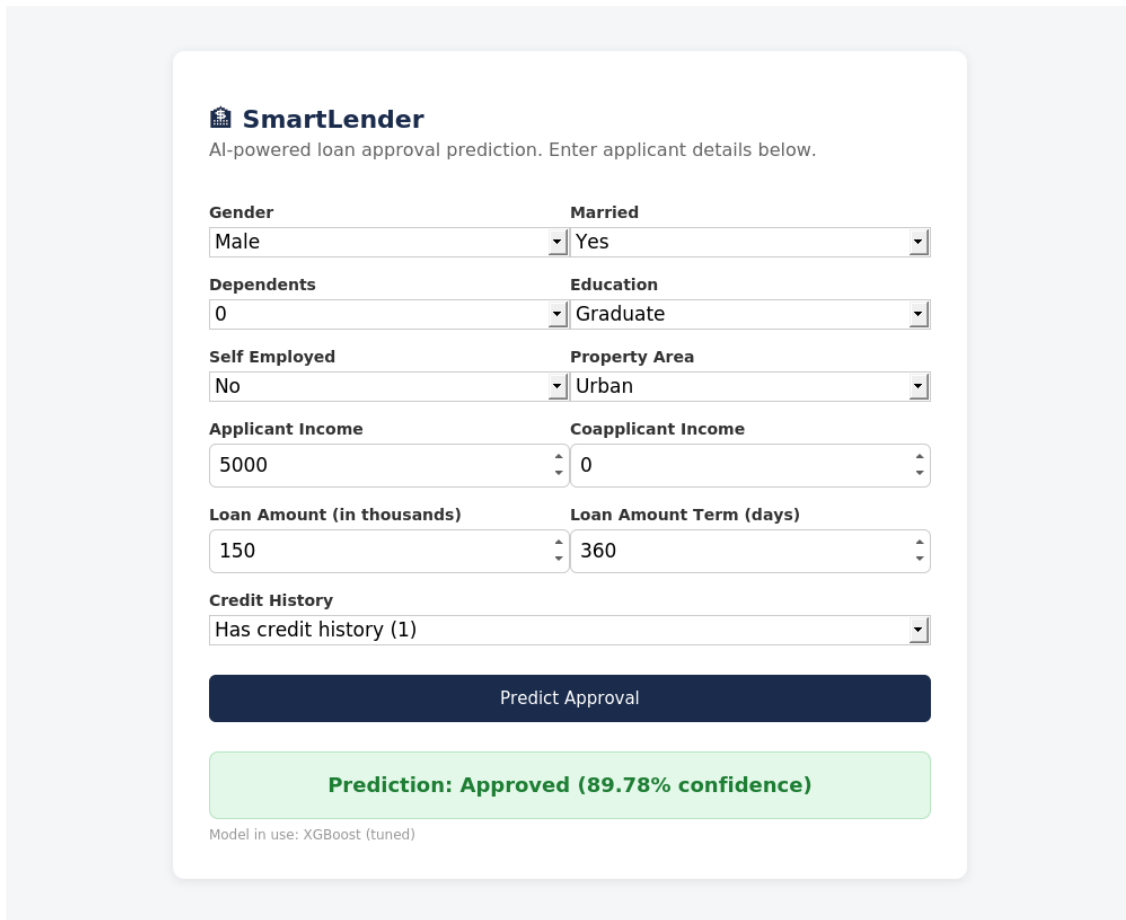
Predict Approval

Model in use: XGBoost (tuned)

Figure 6: Input form interface

Web Interface – Prediction Output

After submission, the system returns the approve/reject decision along with the tuned XGBoost model's confidence percentage.



The image shows a web interface for 'SmartLender', an AI-powered loan approval prediction system. The interface is a light gray card with a white background. At the top, it has the 'SmartLender' logo and the text 'AI-powered loan approval prediction. Enter applicant details below.' Below this, there are several input fields arranged in two columns. The first column contains 'Gender' (Male), 'Dependents' (0), 'Self Employed' (No), 'Applicant Income' (5000), 'Loan Amount (in thousands)' (150), and 'Credit History' (Has credit history (1)). The second column contains 'Married' (Yes), 'Education' (Graduate), 'Property Area' (Urban), 'Coapplicant Income' (0), and 'Loan Amount Term (days)' (360). Each input field is a white box with a dropdown arrow on the right. Below the input fields is a dark blue button labeled 'Predict Approval'. Below the button is a green box with the text 'Prediction: Approved (89.78% confidence)'. At the bottom, there is a small text label 'Model in use: XGBoost (tuned)'.

SmartLender
AI-powered loan approval prediction. Enter applicant details below.

Gender	Married
Male	Yes
Dependents	Education
0	Graduate
Self Employed	Property Area
No	Urban
Applicant Income	Coapplicant Income
5000	0
Loan Amount (in thousands)	Loan Amount Term (days)
150	360
Credit History	
Has credit history (1)	

Predict Approval

Prediction: Approved (89.78% confidence)

Model in use: XGBoost (tuned)

Figure 7: Prediction output display